SAFETY DATA SHEET

According to Regulation (EC) No. 1907/2006 (REACH) Article 31, Annex II as amended 2015/830.

SECTION 1: Identification of the substance/mixture and of the company/undertaking

1.1 Product identifier

Product name: Chloroform

Product No. 5285, 7071, 7152, 7386, 9174, 9180, 4429, 4443, 4444, 5411, 6754

Synonyms, Trade Names: Trichloromethane; Methyl trichloride; Methane trichloride

Additional identification

Chemical name:	Chloroform
Chemical formula:	CHCl3
INDEX No.	602-006-00-4
CAS-No.	67-66-3
EC No.	200-663-8
REACH Registration No.	01-2119486657-20-XXXX

1.2 Relevant identified uses of the substance or mixture and uses advised against

Identified uses: For Laboratory, Research or Manufacturing Use.
Uses advised against: Not determined.

1.3 Details of the supplier of the safety data sheet

Avantor Performance Materials Poland S.A.
Sowinskiego 11str., 44-101 Gliwice,
Poland

Telephone: 48 32 239-20-00
Fax: 48 32 239-23-70

Contact person: Product Information Compliance
E-mail: export@avantormaterials.com

E-mail address of person responsible for this SDS: SDS@avantormaterials.com

1.4 Emergency telephone number: CHEMTREC: (44)-870-8200418

SECTION 2: Hazards identification

2.1 Classification of the substance or mixture

The product has been classified according to the legislation in force.

Classification according to Regulation (EC) No 1272/2008 as amended.

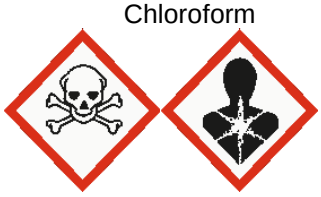
Health Hazards

Acute toxicity (Oral)	Category 4	H302: Harmful if swallowed.
Acute toxicity (Inhalation - vapour)	Category 3	H331: Toxic if inhaled.
Skin irritation	Category 2	H315: Causes skin irritation.
Serious eye irritation	Category 2	H319: Causes serious eye irritation.
Carcinogenicity	Category 2	H351: Suspected of causing cancer.
Toxic to reproduction	Category 2	H361d: Suspected of damaging the unborn child.

Specific Target Organ Toxicity - Repeated Exposure	Category 1	H372: Causes damage to organs through prolonged or repeated exposure.

2.2 Label Elements

Contains: Chloroform



Signal Word: Danger

Hazard Statement(s):
H331: Toxic if inhaled.
H302: Harmful if swallowed.
H315: Causes skin irritation.
H319: Causes serious eye irritation.
H351: Suspected of causing cancer.
H372: Causes damage to organs through prolonged or repeated exposure.

H361d: Suspected of damaging the unborn child.

Precautionary Statements

Prevention:
P201: Obtain special instructions before use.
P260: Do not breathe dust/fume/gas/mist/vapours/spray.
P264: Wash thoroughly after handling.
P280: Wear protective gloves/protective clothing/eye protection/face protection.

Response:
P304+P340: IF INHALED: Remove person to fresh air and keep comfortable for breathing.
P311: Call a POISON CENTER/doctor.
P337+P313: If eye irritation persists: Get medical advice/attention.
P308+P313: IF exposed or concerned: Get medical advice/attention.

Storage:
P403+P233: Store in a well-ventilated place. Keep container tightly closed.

2.3 Other hazards No data available.

SECTION 3: Composition/information on ingredients

3.1 Substances

Chemical name	Concentration	CAS-No.	EC No.	REACH Registration No.	M-Factor:	Notes
Chloroform	50 - <100%	67-66-3	200-663-8	01-2119486657-20-XXXX	No data available.	#

* All concentrations are percent by weight unless ingredient is a gas. Gas concentrations are in percent by volume.
This substance has workplace exposure limit(s).

SECTION 4: First Aid Measures

General:	Get medical advice/attention if you feel unwell. Show this safety data sheet to the doctor in attendance.

4.1 Description of first aid measures

Inhalation:	Move to fresh air. Get medical attention if symptoms occur.
Skin Contact:	Wash skin thoroughly with soap and water. Get medical attention if symptoms occur. Wash contaminated clothing before re-use.
Eye contact:	Immediately flush with plenty of water for at least 15 minutes. If easy to do, remove contact lenses. Call a doctor or poison control centre immediately.
Ingestion:	Rinse mouth. Call a POISON CENTRE/doctor if you feel unwell.
4.2 Most important symptoms and effects, both acute and delayed:	Harmful if swallowed. Irritating to eyes, respiratory system and skin. Narcotic effect. May cause reproductive effects.

4.3 Indication of any immediate medical attention and special treatment needed

Hazards:	No data available.
Treatment:	Treat symptomatically. Symptoms may be delayed.

SECTION 5: Firefighting Measures

General Fire Hazards:	In case of fire and/or explosion do not breathe fumes.

5.1 Extinguishing media

Suitable extinguishing media:	Water spray, foam, dry powder or carbon dioxide.
Unsuitable extinguishing media:	None known.
5.2 Special hazards arising from the substance or mixture:	Fire may produce irritating, corrosive and/or toxic gases.

5.3 Advice for firefighters

Special fire fighting procedures:	Move containers from fire area if you can do so without risk. Use water spray to keep fire-exposed containers cool. Cool containers exposed to flames with water until well after the fire is out.
Special protective equipment for firefighters:	Firefighters must use standard protective equipment including flame retardant coat, helmet with face shield, gloves, rubber boots, and in enclosed spaces, SCBA. Self-contained breathing apparatus and full protective clothing must be worn in case of fire.

SECTION 6: Accidental Release Measures

6.1 Personal precautions, protective equipment and emergency procedures:	Keep unauthorised personnel away. Ventilate closed spaces before entering them. Use personal protective equipment. See Section 8 of the SDS for Personal Protective Equipment. Do not touch damaged containers or spilled material unless wearing appropriate protective clothing.
6.2 Environmental Precautions:	Do not contaminate water sources or sewer. Prevent further leakage or spillage if safe to do so. Avoid discharge into drains, water courses or on to the ground.

6.3 Methods and material for containment and cleaning up: Absorb spill with vermiculite or other inert material, then place in a container for chemical waste. Clean surface thoroughly to remove residual contamination. Dyke far ahead of larger spill for later recovery and disposal.

6.4 Reference to other sections: No data available.

SECTION 7: Handling and Storage:

7.1 Precautions for safe handling: Use personal protective equipment as required. Avoid breathing mists or vapours. Avoid contact with eyes, skin, and clothing. Do not taste or swallow. Wash hands thoroughly after handling. Do not handle until all safety precautions have been read and understood. Obtain special instructions before use.

7.2 Conditions for safe storage, including any incompatibilities: Keep containers tightly closed. Keep in a cool, well-ventilated place. Store in a dry place.

7.3 Specific end use(s): No data available.

SECTION 8: Exposure Controls/Personal Protection

8.1 Control Parameters

Occupational Exposure Limits

Chemical name	Type	Exposure Limit Values		Source
Chloroform	TWA	2 ppm	9,9 mg/m3	UK. EH40 Workplace Exposure Limits (WELs) (2007)
	TWA	2 ppm	10 mg/m3	EU. Indicative Exposure Limit Values in Directives 91/322/EEC, 2000/39/EC, 2006/15/EC, 2009/161/EU (12 2009)

8.2 Exposure controls

Appropriate Engineering Controls: No data available.

Individual protection measures, such as personal protective equipment

General information: Good general ventilation (typically 10 air changes per hour) should be used. Ventilation rates should be matched to conditions. If applicable, use process enclosures, local exhaust ventilation, or other engineering controls to maintain airborne levels below recommended exposure limits. If exposure limits have not been established, maintain airborne levels to an acceptable level. An eye wash and safety shower must be available in the immediate work area.

Eye/face protection: Wear safety glasses with side shields (or goggles) and a face shield.

Skin protection

Hand Protection: Material: Chemical resistant gloves

Other: Wear suitable protective clothing.

Respiratory Protection: In case of inadequate ventilation, use air-supplied full-mask.

Hygiene measures: Provide eyewash station and safety shower. Always observe good personal hygiene measures, such as washing after handling the material and before eating, drinking and/or smoking. Routinely wash work clothing to remove contaminants. Discard contaminated footwear that cannot be cleaned.

Environmental Controls: No data available.

SECTION 9: Physical And Chemical Properties

9.1 Information on basic physical and chemical properties

Appearance

Physical state:	Liquid
Form:	Liquid
Colour:	Colourless
Odour:	sweetish
Odour Threshold:	No data available.
pH:	No data available.
Freezing point:	-63,41 °C
Boiling Point:	61 °C
Flash Point:	No data available.
Evaporation Rate:	11,6 (butyl acetate=1)
Flammability (solid, gas):	No data available.
Flammability limit - upper (%)	No data available.
Flammability limit - lower (%)	No data available.
Vapour pressure:	26,3 kPa (25 °C)
Vapour density (air=1):	4,12 Air=1
Density:	1,48 g/ml (20 °C)
Relative density:	1,48 (20 °C)
Solubility(ies)	
Solubility in Water:	5 g/l (25 °C)
Solubility (other):	Alcohol: Miscible
Partition coefficient (n-octanol/water):	1,97
Autoignition Temperature:	No data available.
Decomposition Temperature:	No data available.
Viscosity:	No data available.
Explosive properties:	No data available.
Oxidising Properties:	No data available.

9.2 Other information

Molecular weight:	119,38 g/mol (CHCl3)
VOC content:	EC Directive 2004/42: 1.000 g/l ~100 % (calculated)

SECTION 10: Stability and Reactivity

10.1 Reactivity: No dangerous reaction known under conditions of normal use.

10.2 Chemical Stability: Material is stable under normal conditions.

10.3 Possibility of Hazardous Reactions: Hazardous polymerization does not occur.

10.4 Conditions to Avoid: Heat, sparks, flames. Contact with incompatible materials.

10.5 Incompatible Materials: Strong oxidising agents. Strong bases. Caustics. Aluminium. Chemically active metals.

10.6 Hazardous Decomposition Products: Oxides of Carbon. Hydrogen Chloride. Chlorine.

SECTION 11: Toxicological Information

Information on likely routes of exposure

Inhalation: Toxic if inhaled.

Skin Contact: Causes skin irritation.

Eye contact: Causes serious eye irritation.

Ingestion: Harmful if swallowed.

11.1 Information on toxicological effects

Acute toxicity

Oral
Product: LD 50 (Rat): 444 mg/kg

Dermal
Product: No data available.

Inhalation
Product: LC 50 (Rat, 4 h): 47,702 mg/l

Repeated dose toxicity
Product: No data available.

Skin Corrosion/Irritation:
Product: Causes irritation.

Serious Eye Damage/Eye Irritation:
Product: Causes serious eye irritation.

Respiratory or Skin Sensitisation:
Product: Not a skin sensitiser.

Germ Cell Mutagenicity

In vitro
Product: No mutagenic components identified

In vivo
Product: No mutagenic components identified

Carcinogenicity
Product: Suspected of causing cancer.

Reproductive toxicity
Product: Suspected of damaging fertility or the unborn child.

Specific Target Organ Toxicity - Single Exposure
Product: Central nervous system.

Specific Target Organ Toxicity - Repeated Exposure
Product: Liver. Kidneys.

Aspiration Hazard
Product: Not classified

Other Adverse Effects: None known.

SECTION 12: Ecological Information

12.1 Toxicity

Acute toxicity

Fish
Product: No data available.

Specified substance(s)
Chloroform — LC 50 (Fathead minnow (Pimephales promelas), 96 h): 62,81 - 79,61 mg/l (Flow through)
LC 50 (Bluegill (Lepomis macrochirus), 96 h): 18,3 mg/l (Flow through)

Aquatic Invertebrates
Product: No data available.

Specified substance(s)
Chloroform — EC 50 (Brine shrimp (Artemia salina), 24 h): 28,6 - 33,8 mg/l (Static)
EC 50 (Water flea (Daphnia magna), 24 h): 79 mg/l (Static)
LC 50 (Water flea (Daphnia magna), 216 h): 200 - 512 mg/l (Static)
LC 50 (Water flea (Daphnia magna), 48 h): 19 - 47 mg/l (Static)

Chronic toxicity

Fish
Product: No data available.

Specified substance(s)
Chloroform — No data available.

Aquatic Invertebrates
Product: No data available.

Specified substance(s)
Chloroform — No data available.

Toxicity to aquatic plants
Product: No data available.

Specified substance(s)
Chloroform — No data available.

12.2 Persistence and Degradability

Biodegradation
Product: The product is not readily biodegradable.

Specified substance(s)
Chloroform — No data available.

BOD/COD Ratio
Product No data available.

Specified substance(s)

Chloroform	No data available.

12.3 Bioaccumulative Potential

Product:	No data available on bioaccumulation.
Specified substance(s)	
Chloroform	Bluegill (Lepomis macrochirus), Bioconcentration Factor (BCF): 6 (Flow through)

12.4 Mobility in Soil:	No data available.
Known or predicted distribution to environmental compartments	
Chloroform	No data available.
12.5 Results of PBT and vPvB assessment:	Not available.
Chloroform	No data available.
12.6 Other Adverse Effects:	Harmful to aquatic life with long lasting effects.
12.7 Additional Information:	No data available.

SECTION 13: Disposal Considerations

13.1 Waste treatment methods

General information:	No data available.
Disposal methods:	Discharge, treatment, or disposal may be subject to national, state, or local laws.

SECTION 14: Transport Information

ADR

14.1 UN Number:	UN 1888
14.2 UN Proper Shipping Name:	CHLOROFORM
14.3 Transport Hazard Class(es)	
Class:	6.1
Label(s):	6.1
Hazard No. (ADR):	60
Tunnel restriction code:	(E)
14.4 Packing Group:	III
14.5 Environmental Hazards:	No
14.6 Special precautions for user:	Not determined.

RID

14.1 UN Number:	UN 1888
14.2 UN Proper Shipping Name	CHLOROFORM
14.3 Transport Hazard Class(es)	
Class:	6.1
Label(s):	6.1
14.4 Packing Group:	III
14.5 Environmental Hazards:	No
14.6 Special precautions for user:	Not determined.

IMDG

14.1 UN Number:	UN 1888
14.2 UN Proper Shipping Name:	CHLOROFORM
14.3 Transport Hazard Class(es)	
Class:	6.1
Label(s):	6.1
EmS No.:	F-A, S-A
14.4 Packing Group:	III
14.5 Environmental Hazards:	No
14.6 Special precautions for user:	Not determined.

IATA

14.1 UN Number:	UN 1888
14.2 Proper Shipping Name:	Chloroform
14.3 Transport Hazard Class(es):	
Class:	6.1
Label(s):	6.1
14.4 Packing Group:	III
14.5 Environmental Hazards:	No
14.6 Special precautions for user:	Not determined.

14.7 Transport in bulk according to Annex II of MARPOL and the IBC Code: Not applicable

SECTION 15: Regulatory information

15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture:

EU Regulations

Regulation (EC) No. 2037/2000 Substances that deplete the ozone layer: none

Regulation (EC) No. 850/2004 on persistent organic pollutants: none

Regulation (EC) No. 689/2008 Import and export of dangerous chemicals: none

Regulation (EC) No. 1907/2006 REACH Annex XIV Substance subject to authorisation, as amended: none

Regulation (EC) No. 1907/2006 Annex XVII Substances subject to restriction on marketing and use:

The packaging shall be visibly, legibly and indelibly marked as follows:
For use in industrial installations or professional treatment only.

Chemical name	CAS-No.	Concentration
Chloroform	67-66-3	100%

Directive 2004/37/EC on the protection of workers from the risks related to exposure to carcinogens and mutagens at work.: none

Directive 92/85/EEC: on the safety and health of pregnant workers and workers who have recently given birth or are breast feeding.:

Chemical name	CAS-No.	Concentration
Chloroform	67-66-3	100%

Directive 96/82/EC (Seveso III): on the control of major accident hazards involving dangerous substances: none

EU. Regulation No. 166/2006 PRTR (Pollutant Release and Transfer Registry), Annex II: Pollutants:

Chemical name	CAS-No.	Concentration
Chloroform	67-66-3	100%

Directive 98/24/EC on the protection of workers from the risks related to chemical agents at work:

Chemical name	CAS-No.	Concentration
Chloroform	67-66-3	100%

15.2 Chemical safety assessment: No Chemical Safety Assessment has been carried out.

SECTION 16: Other Information

Revision Information: Not relevant.

References

PBT	PBT: persistent, bioaccumulative and toxic substance.
vPvB	vPvB: very persistent and very bioaccumulative substance.

Key literature references and sources for data: No data available.

Wording of the H-statements in sections 2 and 3

H302	Harmful if swallowed.
H315	Causes skin irritation.
H319	Causes serious eye irritation.
H331	Toxic if inhaled.
H351	Suspected of causing cancer.
H361	Suspected of damaging fertility or the unborn child.
H361d	Suspected of damaging the unborn child.
H372	Causes damage to organs through prolonged or repeated exposure.

Training information: No data available.

Issue Date: 17.10.2018
SDS No.:

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